

Introduction to Cryptocurrencies, Fundamental and Technical Analysis

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Trading Patterns, Techniques and Set-Ups

This section introduces core tools for analysing price action, identifying patterns and planning trades using structured and repeatable methods. The focus is on logic, probabilities and clarity (so not prediction).

It also covers trading psychology, including risk management, discipline, and emotional control. The goal of the notes is to help build consistent, adaptable strategies in response to changing market conditions.

Foundations of Technical Analysis (TA)

This section introduces the visual language of charts. It teaches how to interpret price action and identify structures that form the basis of trade ideas. This is the essence of technical analysis: studying price itself.

Chart Types

Candlestick and bar charts are the most common visual formats. Each candle or bar shows four key price points over a time period: Open, High, Low, Close (OHLC).

Candle Shapes and Meaning

Different candle patterns offer insights into market sentiment:

- Doji: indecision — open and close are nearly the same
- Hammer: potential bullish reversal after a downtrend
- Engulfing: strong shift in control between buyers and sellers (Bullish and Bearish)



Identifying trends

Understanding how to define a trend using higher highs and higher lows (uptrend) or lower highs and lower lows (downtrend). Also learning when a market is range-bound or consolidating.

Support and resistance

Resistance: a level where price tends to stop rising and may reverse downward (sellers appear).

Trend Structure and Market Shifts

A strong uptrend usually means **higher highs and higher lows**. But when price stops making new highs or breaks below the previous low which is a sign the trend might be weakening or even flipping. Same goes for downtrends (lower highs and lower lows).

Entry and exit zones

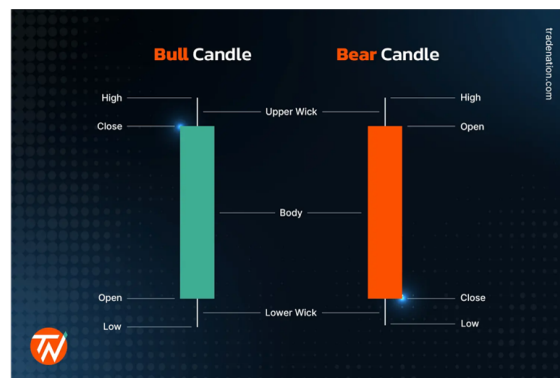
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This section will involve walking through a real chart (e.g. BTC/USDT on TradingView) and will show how to break down its key elements. The aim is to introduce the foundational tools and visual cues used in TA.

- Entry Zones should Include: Support/Resistance, Strong Volume, Indicator Confluence



Understanding Candlesticks



Each candlestick represents price movement over a fixed time period (e.g. 1H, 4H, 1D). The candle shows:

Open: where the price started

Close: where the price ended

High: the highest point reached

Low: the lowest point reached

If the candle closed higher than it opened, it's shown in green (bullish). If it closed lower, it's red or orange meaning bearish.

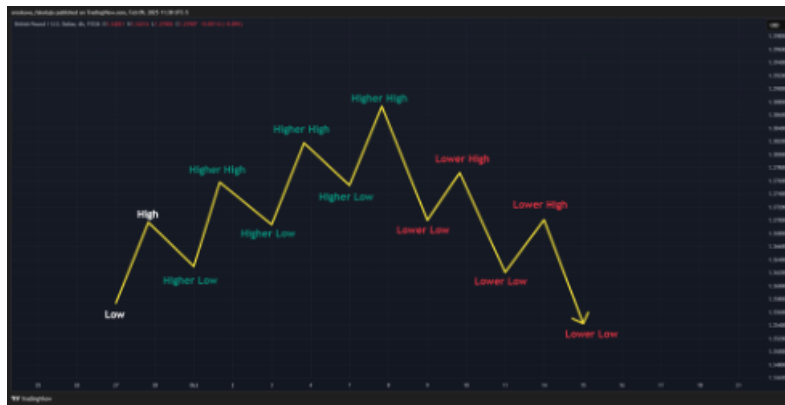
This tells us about market sentiment in that period: were buyers in control? – then green, or were the sellers? – then red.

Market Structure: Higher Highs and Lower Lows

One of the first things traders look for is the structure of the market: is it trending or ranging? (trend is your friend!)

- **Uptrend:** a series of higher highs (HH) and higher lows (HL)
- **Downtrend:** a series of lower highs (LH) and lower lows (LL)

This helps define the trend and identify potential areas to enter (on a higher low in an uptrend, for e.g).



Range Bound



ETH was range bound between \$2200 and \$2800 for 3 months repeatedly touching support and resistance. It eventually broke out after repeated taps of resistance ceiling.

Introducing the Simple Moving Average (SMA)

The SMA is a line that shows the average price over a set number of candles (e.g. 20-period SMA). It **smooths out price action** and helps define the trend. If price is above the SMA, we're generally in an uptrend. If price is below, the trend may be bearish

Below we see 3 different Moving Averages. 50D, 100D and 200D. They typically operate as a "line in the sand" indicating trend and momentum. The longer the average, the more important it is to stay above. Certain moving average pairs

work very well together. 50D and 200D Moving averages when converging to the upside are called golden cross and vice versa is called dooms cross. Markets tend to pump into golden crosses (capitulate on crossing), and crash into doom crosses (rally on/post cross). These are indicated below with circles.



Reading Volume

Volume shows how much of the asset is being traded. It confirms the strength or weakness of a move.

- A breakout with high volume is stronger than one with low volume.
- If volume is dropping during a trend, it could be losing momentum.
- Spikes in volume often occur before or during major moves.

Use the volume bars under the chart to point out:

- A volume spike on a breakout candle
- Low volume during sideways chop



Notice the breakout in November 2024. The price pushed through previous resistance with a strong bullish candle. What confirmed the breakout wasn't just the price move but it was the spike in volume.

High volume during breakouts = strong conviction from buyers!

Following this breakout, Bitcoin ran up nearly 30% in just a few weeks, showing how volume can help validate moves and spot momentum early.

Additional Candles – Heikin Ashi

Beyond standard Japanese candlesticks, traders often use Heikin Ashi (HA) candles to filter out market noise. HA candles use averaged price data, which makes trends easier to spot.

Unlike regular candlesticks, HA candles smooth out price action by using averaged price data, which helps filter out market noise and highlight trends more clearly.

Here's how the values are calculated:

- $HA\ Close = (Open + High + Low + Close) \div 4$
 - This gives the average price of the candle.
- $HA\ Open = (Previous\ HA\ Open + Previous\ HA\ Close) \div 2$
 - A smoothed midpoint from the last candle, not the actual open.
- $HA\ High = \text{Max}(High, HA\ Open, HA\ Close)$
- $HA\ Low = \text{Min}(Low, HA\ Open, HA\ Close)$

However, they lag slightly behind actual price action due to the smoothing effect.

Benefits of HA candles: Smoother charts and clearer visualisation of trend direction.

Limitations: Because prices are averaged, HA candles do not always reflect the actual market price at that time.

Comparing BTC/USDT on the same time frame using standard candlesticks vs HA

The HA chart makes trends look cleaner and easier to spot... but at the cost of lag.

If you only buy once a HA trend confirms, you will avoid chop. However you often miss part of the move. It's slower, but more reliable. Trade-off between certainty vs early entry.

Below we see that HA candles can miss the initial upside movement by about 1 month typically. This can mean a lot of gains lost in a fast moving markets. On the below time frames we would miss out on gains varying from %2 - %40.

Monthly Normal Candle



Monthly HA Candle



Weekly Normal Candle



Weekly HA Candle



As we can see again, price action is smoothed out, and trends become clearer. However, trading of HA can result in cost of missing out on initial rally.

Consecutive green candles = strong uptrend

Smaller or wicking candles = possible slowdown or reversal

But - Because HA candles use averaged data e.g. open = average of previous candle, they lag behind real price. This means:

- Entry signals come later, so you might miss the early part of an uptrend
- Exit signals are delayed, which can cost profits on sharp reversals

TL;DR: HA is great for clarity and trend confirmation, but not for fast reactions.

Bull Market Support Band

A widely referenced long-term indicator (popularized by analyst Ben Cowen) is the Bull Market Support Band, made up of 20-week SMA and 21-week EMA (Exponential Moving Average).

If price stays above this band → market is generally bullish.

If price breaks below and fails to reclaim → market may shift bearish.

BTC historical chart from 2017 or 2020–2021, highlighting how the bull market support band acted as dynamic support during rallies and as resistance in bear phases.



Bull Market Support Band operates as an indicator whether market sentiment is bullish or bearish. Breakouts with confirmation bounces tend to lead to rallies. These are highlighted by black circles.

Key Indicators and Tools

This section introduces commonly used technical indicators that help traders assess market strength, momentum, and potential turning points. These tools are not used in isolation and so emphasis is placed on confluence, where multiple signals reinforce a trade idea.

RSI (Relative Strength Index)

- RSI measures the speed and magnitude of recent price changes to show whether an asset is overbought or oversold.
- A reading above 70 typically suggests overbought conditions (possible reversal or slowdown).
- A reading below 30 suggests oversold conditions (possible bounce or recovery).
- Divergence happens when price makes higher highs but RSI makes lower highs (or vice versa), often signalling a weakening trend.

MACD (Moving Average Convergence Divergence)

- A momentum indicator built from two moving averages (usually the 12-EMA and 26-EMA).
- MACD line crossing above the signal line can suggest bullish momentum.
- Crossing below can indicate bearish momentum.

Moving Averages (SMA & EMA)

- Moving Averages smooth out price data to help identify the direction of the trend.

- SMA (Simple Moving Average) gives equal weight to all prices over a period.
- EMA (Exponential Moving Average) gives more weight to recent prices, making it react faster.
- They are commonly used for trend confirmation, dynamic support/resistance, and crossovers (e.g. 50 EMA crossing above 200 EMA = potential bullish signal).

Volume

- Volume shows how much trading is happening.
- A volume spike can confirm the strength of a breakout or breakdown.
- Low volume during a move may suggest it's weak or likely to reverse.
- Strong moves supported by strong volume are generally more trustworthy.

Confluence

- Good trading setups often combine multiple signals.
- Using 2–3 confirming indicators together increases conviction and filters out false signals.

Example of Confluence in Action:

- RSI Divergence: Price makes lower lows, but RSI makes higher lows → potential bullish reversal.
- Key Support Level: Price is approaching a well-tested support zone from a higher timeframe.
- Candlestick Confirmation: A bullish engulfing candle prints at support.
- Volume Spike: Higher-than-average volume confirms strong participation.

Taken together, these signals provide stronger evidence of a possible reversal than any one of them alone!!

Lagging vs Leading Indicators

Lagging Indicators: These follow price action and confirm trends after they have already started. Examples include Moving Averages, MACD, Bollinger Bands. They are useful for confirmation but may give signals too late, especially in fast-moving markets.

Leading (Predictive) Indicators: These attempt to forecast future price movements before they happen. Examples include RSI, Stochastic Oscillator - compares closing price to recent range to signal potential reversals and Fibonacci retracements – uses key levels based on the Fibonacci sequence to identify possible support and resistance zones.

While they can give early signals, they also carry a higher risk of false positives.

Risks:

- Relying only on lagging indicators can mean entering too late, after the majority of the move has already played out.
- Relying only on leading indicators can result in false signals and overtrading.
- This is why confluence is critical: blending lagging indicators (trend confirmation) with leading indicators (early signals) creates a more balanced and disciplined approach.

Bollinger Bands:

Bollinger Bands (BBs) measure volatility by plotting two standard deviations above and below a moving average (usually 20 SMA).

- When price touches the upper band, market may be **overbought**.
- When price touches the lower band, it may be **oversold**.
- Periods of tight bands (“squeezes”) often precede major breakouts.

In trending markets, BBs can act as a trend confirmation tool. A common example is during late 2020/early 2021 where BTC broke above the upper band and held above the mid-band (20 SMA), leading to a strong upward rally often referred to as riding the bull market support band.

Below is weekly candle price action related to BB. The Top bands typically show markets as overheated, and bottom channel as oversold.



Visual Examples of RSI, MACD and Volume Spikes

For each indicator introduced (RSI, MACD, Bollinger Bands, Volume), include screenshots showing how they looked during an actual BTC/ETH move.

MACD Crossover

As seen from the weekly BTC graph, MACD can operate as a useful indicator in spotting changing trends.



The first and last MACD cross-overs seen highlight certain inefficiencies with MACD. It is a lagging indicator, and prematurely guesses a move, or indicates it too late. For the middle two cross overs, the MACD worked great. Hence the need for confluence.

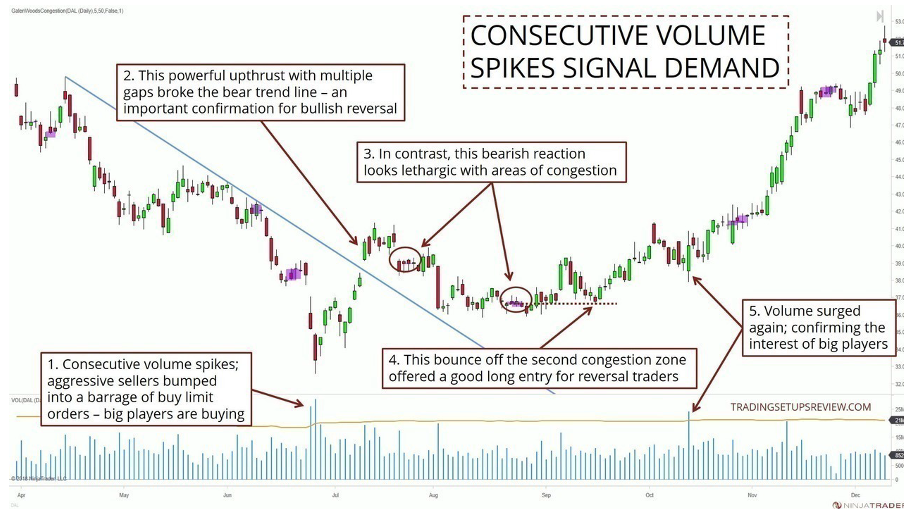
Divergence on RSI



The above chart highlights certain points RSI correctly called bottoms, with RSI_i30, such as the example at the beginning of the chart as well as tops, such as the final RSI peak at the end of the chart. The blue circles highlight points where the

RSI prematurely called tops and bottoms. Once again, highlighting the need for confluence.

Volume Spikes



Disclaimer on Developing Trading Style

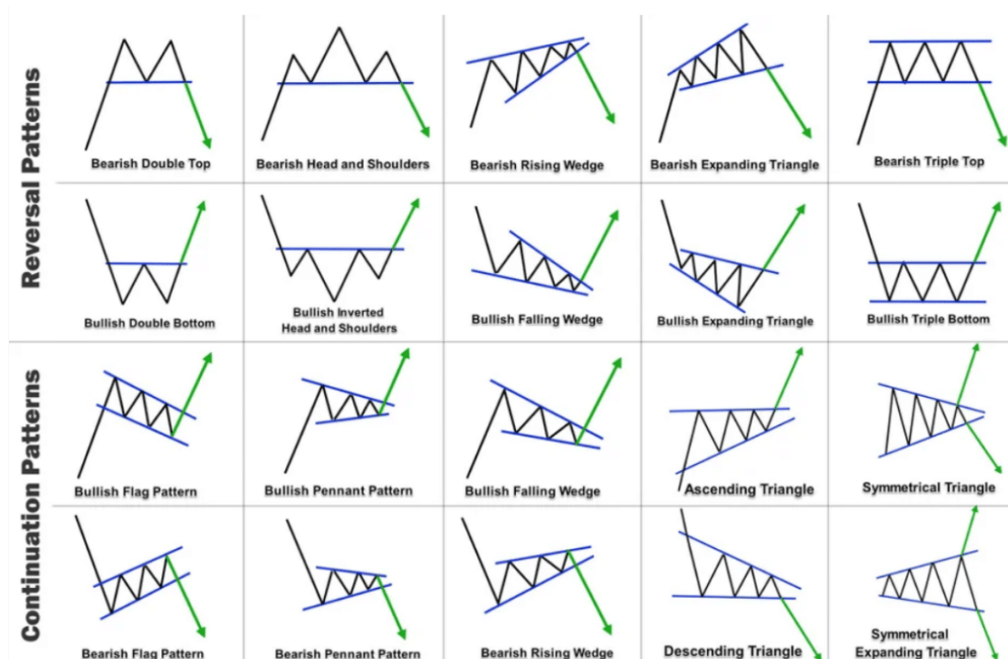
Indicators and patterns are tools, not predictions. No single setup guarantees success. Each trader must build their own style by combining tools, journaling results, and adapting to market conditions. The goal is not copying someone else's system but developing your own framework through practice and reflection.

Common Trading Patterns

Recognising chart patterns is a skill that comes from visual familiarity and practice. Students will study key continuation and reversal patterns and learn how to apply them in context.

Continuation patterns: Bull flags, bear flags, pennants and triangles signal potential trend continuation after a brief pause or consolidation.

Reversal patterns: Double tops/bottoms, head and shoulders and engulfing candles are all signs that a trend may be reversing.



Breakouts

- A breakout happens when price pushes through a key support or resistance level after building pressure.
- This often signals strong momentum and a shift in control between buyers and sellers.
- Look for a full candle close beyond the level, a clear structure like higher lows and a spike in volume. These confirm it's a real move, not just noise.

Fakeouts and Traps

- Fakeouts happen when price breaks a level briefly, then snaps back. This usually traps traders who jumped in too early.
- Avoid this by waiting for proper confirmation and not just a wick. Check for strong volume and see if price retests the level before continuing. This helps filter real moves from false signals.

Trade Setups & Execution

Structured trading means taking your analysis and turning it into an actionable plan. Instead of reacting to the market in the moment, you predefine your conditions for entry, stop loss and take profit. This discipline helps remove emotion from decision-making.

Building a Trading Thesis:

Every trade should start with a clear thesis: why are you taking the trade? To build this thesis, combine three key elements:

- Context: What is the market doing overall? Is it trending, ranging, or consolidating?
- Narrative: What story is the chart telling? For example, “BTC has rejected resistance three times, but RSI is showing bullish divergence at support.”
- Indicators & Levels: Support and resistance zones, trendlines, or confluence with indicators like RSI/MACD. These provide the technical backbone for your trade idea.

Without a thesis, you’re just guessing. With one, you have a framework to measure if the trade is valid.

Entry Points

- Entries are where theory meets execution. Some common methods:
- Breakouts: Entering when price breaks above resistance or below support, ideally confirmed with volume.
- Retests: Waiting for price to come back and retest a broken level before entering — this can provide better risk/reward.
- Candlestick Triggers: Looking for clear reversal candles (e.g. bullish engulfing at support, bearish engulfing at resistance) to confirm timing.

The goal is not to predict every move but to define conditions that must be met before you enter.

Stop Loss Placement

Stop losses protect you when the market proves your thesis wrong. Place them based on technical invalidation, not emotions. For example:

- If you buy at support, your stop goes below that support.
- If you short at resistance, your stop goes above that resistance.

You can also use volatility tools like the ATR (Average True Range) to avoid placing stops too tight, which can often result in being stopped out by normal price noise.

Take Profit Targets

Profits should be taken logically, not emotionally. Use:

- Risk-to-Reward Ratios (R:R): Aim for setups where the potential reward is at least 2x the risk (e.g., risking \$100 to make \$200).
- Support/Resistance: If you long from support, your target could be the next resistance zone.
- Trailing Stops: Instead of fixed targets, trail your stop behind the move to lock in profits while allowing the trend to run.

Comparing these approaches helps you understand when to secure profits and when to let winners ride.

Journaling Trades

Every trade is a chance to learn. Journaling means recording:

- Your entry, stop loss, and take profit.
- Why you entered (your thesis).
- How you felt during the trade.

Over time, patterns emerge basically meaning one should be able to spot which setups work best for them and which emotional triggers lead to mistakes. A trading journal is one of the fastest ways to improve consistency.

Trading Psychology and Mindset

Technical skills are just one part of becoming a successful trader. Even the best trade setups can fail if emotions take over. What separates consistent traders from impulsive ones is mindset as well as how one manages stress, risk and decision-making in uncertain conditions.

- **Patience**

The market doesn't always offer good setups. Rushing in out of boredom or fear of missing out (FOMO) often leads to bad trades. High-quality trades come to those who wait.

- **Discipline**

Once a plan is made, it must be followed. This means no moving stop losses mid-trade, no chasing green candles, and no breaking your own risk rules.

- **Managing Losses**

Losses are normal they don't mean failure. Taking a small, planned loss is better than holding a losing trade and hoping. Knowing when to exit is as important as when to enter.

- **Emotional Regulation**

Markets trigger emotions like greed, fear, and frustration. The goal is to stay neutral i.e to trade the setup, not the feeling. Recognising these triggers early can stop poor decisions.

- **Risk Control**

Protecting capital is the first priority. Good risk control means risking only a small percentage of your capital per trade (e.g. 1–2%), and not over-leveraging or over-trading.

- **Growth Mindset**

A win or loss in your trades is an opportunity to learn. Mistakes are feedback, not failure. Wins shouldn't make you get overconfident either. Curiosity and adaptability help build long-term skills.

- **Reflection**

Keeping a journal of your trades (including emotions, reasoning, and outcomes) helps spot patterns in your behaviour. Reviewing this regularly is key to self-improvement.

Chart Practice Guide: Applying TA Tools in Real Time

1. Marking Charts

- Start by opening a BTC or ETH chart on TradingView.
- Identify key support and resistance zones. Shows where price has reacted multiple times.
- Draw trendlines connecting recent swing highs and swing lows to define the current market structure.
- Look for developing patterns like bull flags, double tops/bottoms, or triangles.

2. Using RSI and MACD

- Add RSI and MACD indicators to your chart.
- Use RSI to spot when price is overbought/oversold and look for divergence (e.g. price making higher highs while RSI makes lower highs).
- Use MACD to watch for crossovers and momentum shifts using the histogram.

Both tools can help confirm your setups especially when they align.

3. Spotting Breakouts

- Watch how price behaves around key levels.
- A breakout is confirmed when price closes above resistance (or below support) with strong volume.
- Before making a move, wait for a clean retest or confirmation candle as this avoids jumping in on a fake move.

Think of it like pressure building up i.e the more times a level is tested, the more likely it is to break.

4. Simulating Trades

- Pick one setup and walk through how you'd trade it.
- Mark your entry point. ideally do this where the breakout is confirmed.
- Place your stop loss below/above the last swing point or invalidation level.

- Set a target at the next key resistance/support level or use a fixed risk-reward (e.g. 2:1).

Example: “Entry on retest of resistance turned support. RSI shows bullish divergence. Stop goes under previous low. Target is next resistance.”

Repeat this with multiple setups to build repetition.

5. BTC Dominance Check

- Search for the BTC.D chart on TradingView.
- If BTC dominance is rising, capital is flowing into BTC and so altcoins may be underperforming.
- If BTC dominance is falling, traders may be rotating into ETH or smaller-cap tokens.
- Use this as a macro filter to understand overall market sentiment before making trades.

For Direct Examples of Technical Analysis on various currencies this is a good source: <https://www.youtube.com/watch?v=tIGPDwduDPo>